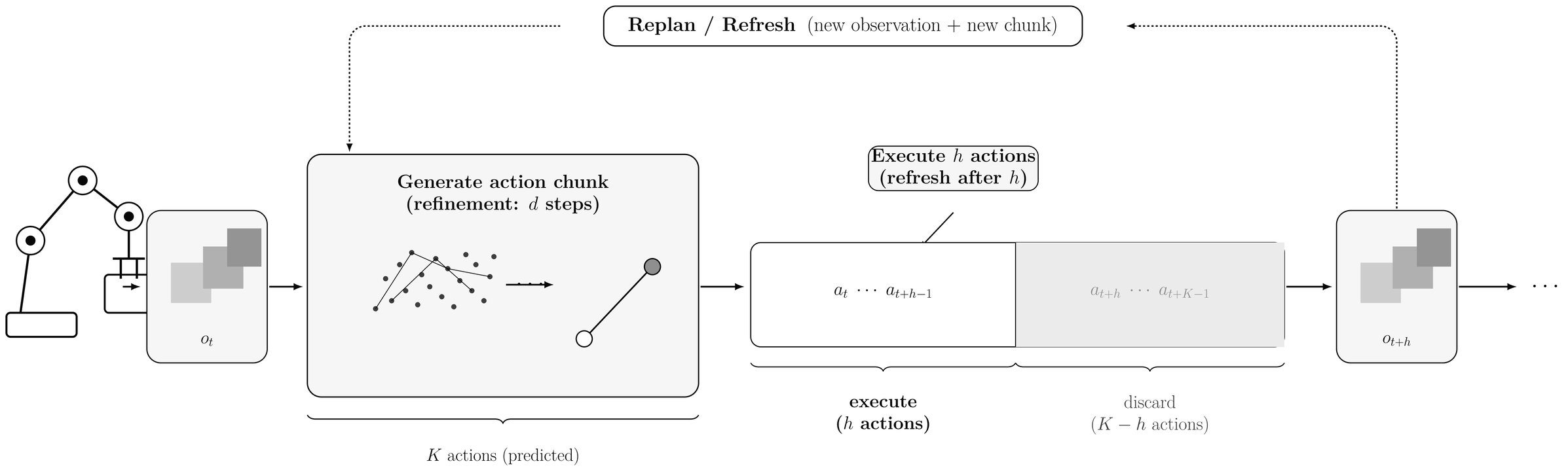
Refinement vs. Refresh

Compute Allocation for Closed-Loop Robot Policies

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Chunked Closed-Loop Policy



K: **chunk length**
(total predicted actions)

d: **refinement**
(denoising steps)

h: **refresh**
(actions before replanning)

Setup (held-out)

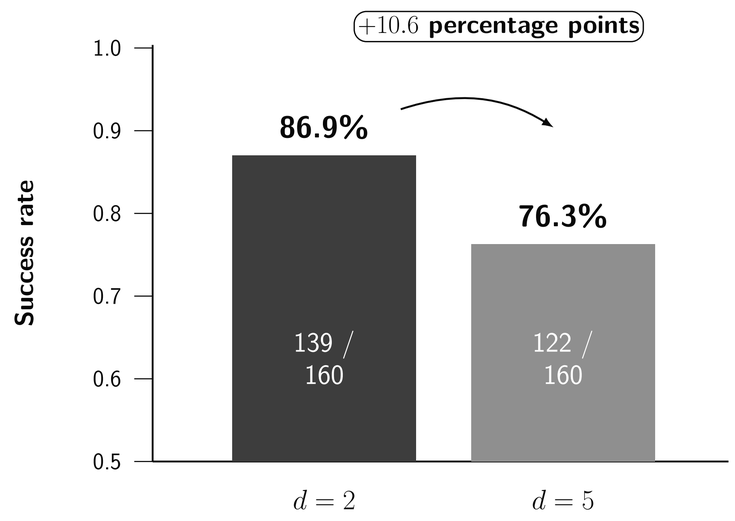
SmolVLA (frozen) $K = 50,\ d \in \{2, 5\}$

LIBERO (simulation) $h \in \{2, 5\}$ 8 tasks, 1,120 attempts

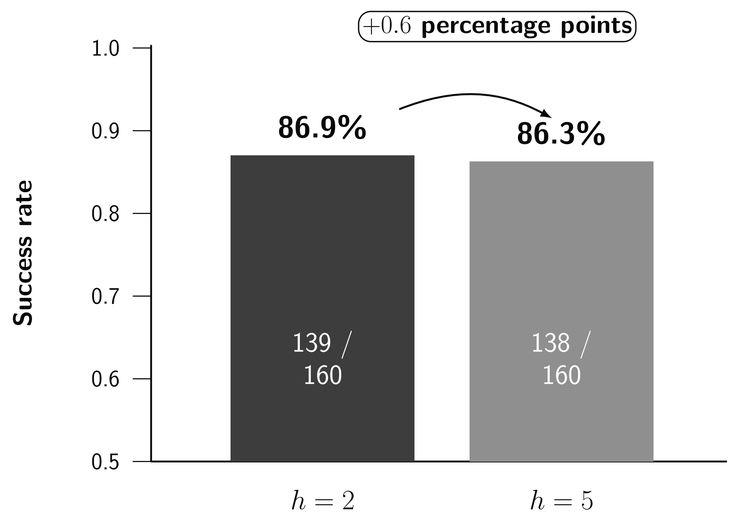
?

Under a fixed inference budget, should we allocate compute to additional denoising steps (**refinement**) or to more frequent observation-conditioned replanning (**refresh**)?

Result 1: Fewer Denoising Steps Improve Success



Result 2: Less Frequent Refresh Is Much Cheaper



What the results mean

Takeaway

Less refinement can improve closed-loop success at lower arithmetic. Longer execution prefixes can be dramatically cheaper, but this comparison remains descriptive.

Interpretation and limits

Together, the results motivate joint refinement-refresh evaluation. They do not establish equivalence, a universal allocation, physical latency, or transfer beyond this checkpoint.